

| Submission                       | Line                                                                              | Score         |
|----------------------------------|-----------------------------------------------------------------------------------|---------------|
| Schedule-Free AdamW v2 (JAX)     |   | <b>0.6852</b> |
| Schedule-Free AdamW v2 (PyTorch) |  | 0.6072        |
| Muon (PyTorch)                   |  | 0.5993        |
| Schedule-Free AdamW (PyTorch)    |  | 0.5786        |
| Schedule-Free AdamW (JAX)        |  | 0.5561        |
| AdEMAMix (PyTorch)               |  | 0.5213        |
| NAdamW (Baseline v0.5) (JAX)     |  | 0.4933        |
| Muon (JAX)                       |  | 0.4506        |
| Lion (PyTorch)                   |  | 0.4439        |
| Cautious NAdamW (JAX)            |  | 0.4012        |
| NAdamW (JAX)                     |  | 0.3754        |
| NAdamW tuned for ResNet (JAX)    |  | 0.3266        |
| Single-worker DiLoCo (JAX)       |  | 0.3199        |
| Single-worker DiLoCo v2 (JAX)    |  | 0.2318        |

